

Assignment 1

1. **Define Cryptography.** In the context of modern digital communication and data security, explain five distinct real-world applications of cryptography. For each application, briefly describe how cryptography is implemented and what specific security goal it achieves (e.g., confidentiality, integrity, authentication, non-repudiation). Discuss one emerging trend or future challenge in the field of cryptography in the modern era.
(15 marks)
2. **Explain the Caesar Cipher and its working.** What are the limitations of this cipher? Provide an example of encrypting a message using the Caesar Cipher.
(15 marks)
3. **Compare and contrast monoalphabetic and polyalphabetic ciphers.** How does the Vigenère Cipher improve upon the Caesar Cipher?
(7 marks)
4. **Describe the Rail Fence Cipher and its working.** Provide an example of encrypting a message using this cipher.
(15 marks)
5. **What is cryptanalysis?** Explain how frequency analysis can be used to break classical substitution ciphers like the Caesar Cipher.
(7 marks)
6. **Explain the different types of intruders and their motivations.** How can an Intrusion Detection System (IDS) help in detecting and preventing unauthorized access? Provide examples of IDS types and their functions.
(7 marks)
7. **What is malicious software (malware)?** Discuss the different types of malwares (e.g., viruses, worms, trojans) and their impact on systems. How do firewalls help in protecting networks from malware and unauthorized access?
(15 marks)
8. **Explain the CIA triad in cryptography and its importance in information security.**
(15 marks)

9. What is the difference between a block cipher and a stream cipher? Provide an example of each.

(15 marks)

10. . What is the purpose of symmetric encryption? Explain the working of the Blowfish algorithm.

(7 Marks)